

Light chaser can be used to create lighting animation sequences. It has been used in the past to draw attention to advertising and promotion; for example, the marquee at the local movie theater. It can also be used to produce pleasing effects for entertainment. Microcontroller-based chasers invaded the fancy lighting pattern and outran the chasers made of fundamental electronic components and logic gates. The potential advantage of microcontroller-based chaser light is that the designer can make LEDs glow in any pattern. But here again problem arises when there are complex steps with step-size of more than a thousand. The case worsens if the delay of each step is different. In such a situation, 'Chaser designer' presented here can play an important role

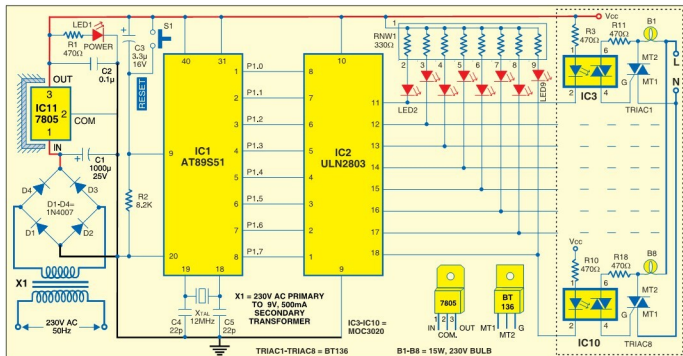
Here we present AT89551 microcontroller-based light chaser. AT89551 is a low-power, high-performance CMOS 8-bit microcontroller with 4KB programmable Flash memory, 128 bytes of RAM, 32 input/output (I/O) lines, watchdog timer, two data pointers, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator, and clock circuitry. System clock plays a significant role in the operation of the microcontroller.

Fig. 1 shows the circuit diagram of a microcontroller-based light chaser. Microcontroller AT89S51 forms the main controller section while ICs MOC3020 and triacs BT136 form the bulb driver section (shown by dotted



A 12MHz quartz crystal at pins 18 and 19 provides clock pulse to the microcontroller (IC1). Power-on reset is provided through electrolytic capacitor C3 and resistor R2. Switch S1 is used for manual reset. Port 1 provides the chasing effect pattern to control the chaser light sequence. Port pins P1.0 through P1.7 are connected to pin 1 of opto-coupler IC3 through IC10, respectively, through IC2 ULN2803. Pin 4 of triac driver MOC3020 provides the gate pulse to the triac.

MOC3020 are optically isolated triac drivers. Each driver contains a gallium arsenide (GaAs) infrared emitting diode and a light-activated silicon bilateral switch, which functions like a triac. They are designed for interfacing



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